



PUNIT POWERCARE

SWITCH TO SUCCESS WITH PUNIT POWERCARE

APFC and Switchgear Care

Flat no. 11 Geetganga Society,
Saptashrungi Nagar,
Old Saykheda Road,
Jail Road, Nashik 422101
Email- punitpowercare@gmail.com

Ganesh Yeole - 9579853838

**Workshop:- MIDC Ambad, Nashik,
Maharashtra 422110**



ABOUT US

24 x 7 **PUNIT POWERCARE** is a well known name in the field of power management and electricity. We are one of the pioneers in developing Thyristorised APFC system.

We are known for quality & after sales service.

Trident Power Enterprises means: Real care of customer and equipments.

WHAT IS POWER FACTOR

Power factor (PF) is the ratio of working power to total power (working power plus non productive reactive power that circulates in your facility).

Technically it is defined as the cosine of the angle between the voltage and the current.

Power factor between 0.95-1.0 indicates maximum utilization of electrical power where as having low PF(<0.9) indicates poor utilization of electric power. Having lower power factor leads to purchasing more power to obtain the same load KW, which you are paying for in various ways as your electricity bill. For Example you may

For Example you may be paying power factor penalties. Reactive power charges or surcharge fee, these are essentially penalties for operating a facility with low power factor

WHY DO WE NEED POWER FACTOR CORRECTION?

The connected equipment (Transformer, Motors, Air conditioners, Furnaces, Rolling mills, Spinning Mills, Variable drives etc.) cause a phase angle between current and voltage when the current is phase shifted, it takes more current to deliver the same amount of active power and we land up paying more bill. This can be avoided by power factor correction.



PUNIT POWERCARE

SWITCH TO SUCCESS WITH PUNIT POWERCARE

Investing in power factor correction is an easy and economically benefitting decision. These systems pay back in 3 to 15 months period depending on the facility load characteristics and the electricity rate structure etc

BENEFIT 1

Reduced Power losses in distribution system. Since losses are proportionate to the square of the current, the formulae:

$$\% \text{ Reduction of power losses} = 100 - \text{Losses} = I^2 R$$

Example Improved PF from 65% to 90%

$$\text{Reduction in power losses} = 100 - 100 \times (.65/90)^2 = 48\%$$

BENEFIT 2

$$KW = KVA \times PF$$

More KW working power for same KVA demand

More KW allows for additional motors or lighting etc.

Example 600 KVA demand Vs available KW

Power Factor	60	70	80	90	100
Active Power(KW)	360	420	480	540	600
Reactive Power(KVAR)	480	428	360	262	0
Total Power(KVA)	600	600	600	600	600

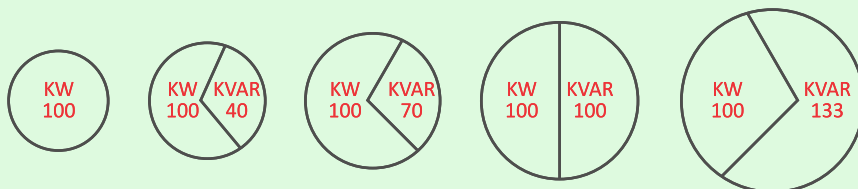
BENEFIT 3

Improved voltage regulation due to reduced line voltage drop. This benefit will result in more efficient performance of motors & other electrical equipment.

BENEFIT 4

$$I = \frac{KVA \times 10^3}{\sqrt{3} V}$$

Reduction in size of Transformer cables and switchgear in new installations thus less investment.



ADVANTAGES OF THYRISTORISED SWITCHED APFC

1. No inrush current due to precise auto zero current search logic.
2. No current or voltage surge. The Transformer is never subjected to short circuit current, Life of the transformer increases.
3. No over compensation.. Improves Voltage Condition.
4. Smooth and instant disconnection of capacitor units. The capacitor bank unit remains in useful operational service with maximum uptime.
5. Solid State devices are designed to work at a low voltage of 200 V, as a result no chattering or sparking during switching of capacitor.
6. The point of capacitor switching is controlled and hence, life of the capacitor unit and switching unit is maximized.
7. Small Steps facilitate close compensation of power factor.
8. The Thyristorised APFC enhances existing load delivering capacity proportionately by maintaining the P.F. to 0.99 or unity.
9. The controller displays various electrical parameters and provides printout facility confirming to RS 232 port to download electrical parameters.
10. Maintenance free, as no moving parts in the system:
11. Short payback period with long term savings.
12. Reduces cable size requirement.

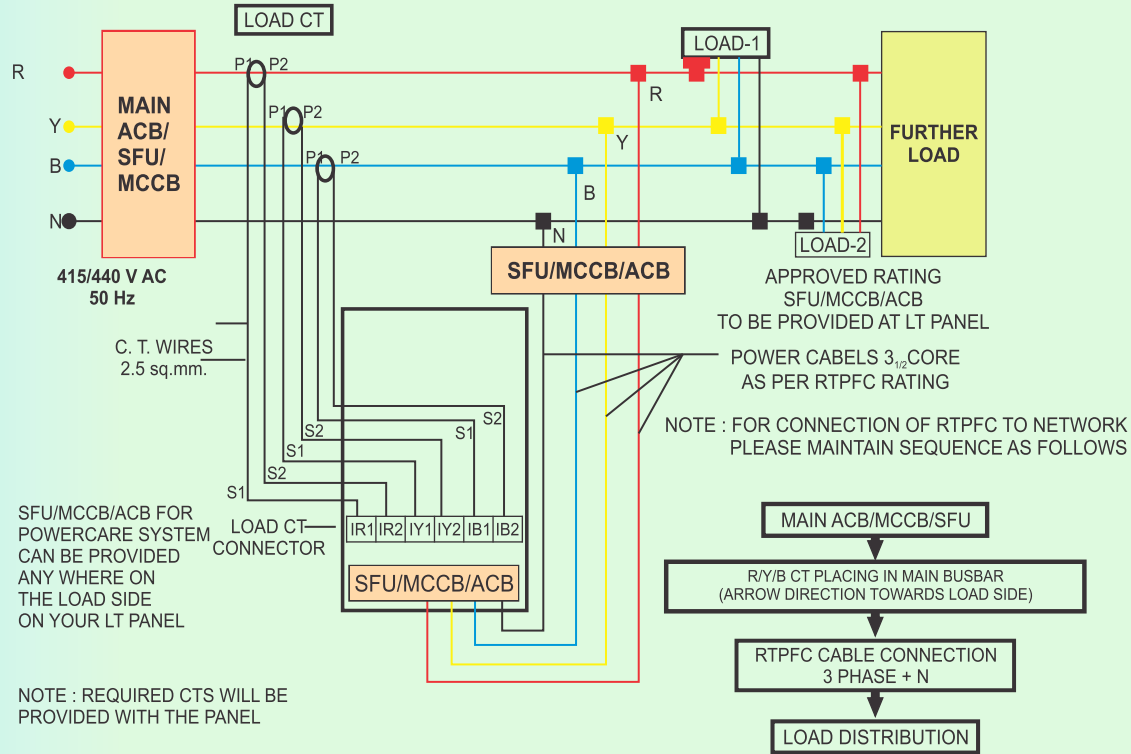


FIXED CAPACITOR VS APFC

Fixed Bank Capacitors may improve the power factor but this may result in over compensation and PF may jump to leading which may involve PF penalty. The electricity bill may also be higher than the actual consumption as the full capacitor bank is always in circuit and is consuming power.

Trident Power Enterprises system solves both the above negatives with economic solution.

CONNECTION DIAGRAM



SERVICE BACK UP

24 x 7 **PUNIT POWERCARE** provides comprehensive support both during the warranty period and after warranty when you enter into AMC With **PUNIT POWERCARE**. Dedicated knowledgeable and experienced staff is available 7 days a week 24 hours a days to answer your questions and dispatch a field service technician to your site.

Useful to :

- Rolling Mills ● Furnaces Load ● Cement Plants ● Steel Plants ● Automobile Industry
- Any Industrial Unit ● Commercial complexes ● Rice Mills ● Stone Crushers ● Hospitals
- Restaurants ● Hotels ● Almost any and every Industry



APFC and Switchgear Care

Flat no. 11 Geetganga Society,
Saptashrungi Nagar,
Old Saykheda Road,
Jail Road, Nashik 422101
Email- punitpowercare@gmail.com

Ganesh Yeole - 9579853838

**Workshop:- MIDC Ambad, Nashik,
Maharashtra 422110**



Our Services:-



- Manufacturing of Contactor based APFC panels.
- Manufacturing of Thyristor based APFC panels.
- Manufacturing and supply of Low tension panels.
(LT PCC, MCC, PDB, PLC, Temperature Control, Fire Panels)
- Supply of servo and relay based voltage stabilizers
(Normal and Odd range).
- Supply of UPS.
- ACB And VCB Servicing With Expert And Well Trained Team
- Annual maintenance contract of APFC panels,
comprehensive and non-comprehensive.
- Annual maintenance contract of LT panels.
- Electrical Panels material supplier.
- leading RKVAH Solution
- AHF, ASVG Supplier



OUR VALUABLE CLIENTS

